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Case 7-2024: A 67-Year-Old Woman with Alternating Sixth Cranial Nerve Palsy

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PRESENTATION OF CASE

Dr. Zandra E. Walton (Medicine): A 67-year-old woman was evaluated at this hospital because of headache and retro-orbital pain.

The patient had been in her usual state of health until 10 months before the current admission, when a photosensitive, erythematous, papular rash developed on the forearms a few weeks after she had started treatment with hydrochlorothiazide. Over the next 7 months, an intermittent, painless, nonpruritic, erythematous, annular rash developed on the trunk and the proximal portions of the arms and legs.

Three months before the current presentation, the rashes worsened, and the patient was evaluated by a dermatologist at another hospital. The thyrotropin level was normal; other laboratory test results are shown in Table 1. Examination of a punch-biopsy specimen of abdominal skin reportedly showed spongiotic dermatitis with superficial perivascular lymphocytic inflammatory-cell infiltrate with eosinophils; neither vasculitis nor dermal mucin was found. The rashes were thought to be related to medication; treatment with amlodipine was started, and hydrochlorothiazide was discontinued. She was treated with topical triamcinolone ointment. The photosensitive rash on the forearms resolved, and the annular rash on the trunk abated.

Ten days after the evaluation at the first hospital, and 8 weeks before the current admission, the patient presented to the emergency department of a second hospital because of progressive temporal headache with retro-orbital pain on the left side that had lasted for 3 days. The physical examination was reportedly normal. The erythrocyte sedimentation rate was 12 mm per hour (reference range, 0 to 19). The complete blood count and blood levels of electrolytes, aspartate aminotransferase, alanine aminotransferase, and total bilirubin were normal, as were the results of tests of kidney function. Computed tomography (CT) of the head was performed and reportedly showed mild periventricular white-matter disease with no acute intracranial abnormality. CT angiography of the head and neck reportedly showed no intracranial aneurysm, stenosis, or occlusion. The patient was discharged home.

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Table 1. Laboratory Data.*

Variable	Reference Range, Adults, First Hospital	3 Mo before Current Admission, First Hospital	Reference Range, Adults, This Hospital†	On Current Admission, This Hospital
Antinuclear antibody	Negative	Positive at 1:320 with a speckled pattern	Negative at 1:80 and 1:160	Positive at 1:640 with a homogeneous pattern
Anti-dsDNA antibody (U)	0–30.0	16.3	Negative	Negative
Anti-RNP antibody (U)	0–20	3	0–19.99	1.48
Anti-Smith antibody (U)	0–20	4	0–19.99	2.74
Anti-SSA antibody (U)	0–20	71	0–19.99	80.76
Anti-SSB antibody (U)	0–20	7	0–19.99	6.69

* The term dsDNA denotes double-stranded DNA, RNP ribonucleoprotein, SSA Sjögren's syndrome–related antigen A, and SSB Sjögren's syndrome–related antigen B.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and therefore may not be appropriate for all patients.

The next morning, the patient awoke with double vision. The headache had resolved. She presented to a neuro-ophthalmology urgent care clinic associated with this hospital. The patient reported persistent pain in the left eye, which was worse with abduction, as well as photosensitivity and an episode of pain in the left side of the jaw in the preceding week. She had normal visual acuity, color vision, visual fields, and afferent visual function. The optic disks were normal, sharp, and flat. The left eye had esotropia and hypertropia, as well as a limitation in abduction. The erythrocyte sedimentation rate was 11 mm per hour, and the C-reactive protein level was 3.0 mg per liter (reference value, <8.0). A presumptive diagnosis of acute left sixth cranial nerve palsy associated with cerebral microvascular disease was considered. Corrective treatment with eye patching was recommended, and the double vision resolved over the next 3 weeks.

Eight weeks after she had presented to the emergency department of the second hospital, the patient presented to the emergency department of a third hospital because similar symptoms had developed in her right eye. She reported having headache with retro-orbital pain on the right side over the past 3 days, accompanied by changes in her color vision and occasional floaters. The physical examination was notable for a blood pressure of 159/66 mm Hg and an abduction deficit of the right eye. The erythrocyte sedimentation rate was 17 mm per hour (reference range, 0 to 30), and the high-sensitivity C-reactive protein

level was 1.64 mg per liter (reference value, <3.00). The complete blood count and blood levels of electrolytes, albumin, aspartate aminotransferase, alanine aminotransferase, and total bilirubin were normal, as were the results of tests of kidney function. A screening test for Lyme disease antibodies was negative.

Dr. Jeremy N. Ford: Cross-sectional imaging of the head and orbits had been performed at the first hospital, and the results were reinterpreted at the third hospital. CT of the head (Fig. 1A and 1B) reportedly showed stranding of the intraconal fat in the right orbit. Magnetic resonance imaging (MRI) and magnetic resonance angiography of the head, neck, and orbits (performed with and without the administration of intravenous contrast material) reportedly showed abnormal enhancement and edema of the optic nerve and surrounding tissue on the right side. On my directed review of the MRI (Fig. 1C through 1F), I found that the mild edema and enhancement predominantly involved the optic-nerve sheath and intraconal fat, which confirmed the findings on CT of the head. The optic nerve itself was largely spared. The temporomandibular joints were partially visible, and there was mild periarthritic edema and enhancement of the left joint. On T2-weighted imaging, scattered hyperintense foci were seen in the periventricular and subcortical white matter, a nonspecific finding most likely attributable to chronic small-vessel disease.

Dr. Walton: Intravenous methylprednisolone was administered, and the patient was transferred to

the neurology unit of this hospital. On evaluation, she reported persistent double vision and pain on right lateral gaze but resolution of the retro-orbital headache. She also reported a recent mild flare of an erythematous, annular rash, as well as recent hair thinning and longstanding Raynaud's phenomenon. Review of systems was negative for fever, night sweats, weight loss, joint erythema or swelling, muscle stiffness or pain, scalp or temple tenderness, jaw claudication, recurrent sinusitis, nasal crusting, epistaxis, cough, hoarseness, cough or shortness of breath, hemoptysis, dry mouth or dry eyes, oral ulcers, ptosis, dysphagia, dysarthria, hearing changes, paresthesia, weakness in the arms and legs, or changes in the nails. There were no gastrointestinal or urinary symptoms.

The patient had a remote history of Graves' disease with orbitopathy that had been diagnosed and treated more than two decades before this presentation. The medical history was also notable for hypothyroidism, hypertension, hyperlipidemia, gastroesophageal reflux disease, cholecystectomy, ovary-sparing hysterectomy, and lumpectomy for estrogen receptor–positive, progesterone receptor–positive breast cancer. A surveillance mammogram obtained in the previous month had reportedly been normal. She had received three vaccines against severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). Medications included atorvastatin, amlodipine, levothyroxine, and tamoxifen. A rash had occurred after the previous use of hydrochlorothiazide. She was married and lived in coastal Massachusetts. She drank wine rarely and did not use tobacco or other substances. Her family history was notable for a maternal aunt with thyroid disease and a child with vitiligo.

On examination, the temporal temperature was 36.1°C, the pulse 68 beats per minute, the blood pressure 173/72 mm Hg, and the oxygen saturation 95% while the patient was breathing ambient air. A few scattered erythematous macules were present on the abdomen, including a well-healed punch-biopsy site. A few telangiectasias were observed on the face. There was no rash on the face or scalp, and no sclerodactyly was observed. The hair was normal in pattern and distribution. There was no conjunctival injection. Extraocular movements were normal except for abduction of the right eye, which was limited and caused pain. There was no proptosis

or lid lag. Visual acuity was 20/25 in the left eye and 20/40 with red color desaturation in the right eye. The right optic disk appeared pale as compared with the left optic disk. The remainder of the general and neurologic examination was normal.

An electrocardiogram was normal. The complete blood count and blood levels of electrolytes, calcium, albumin, and globulin were normal; other laboratory test results are shown in Table 1. The erythrocyte sedimentation rate was 6 mm per hour, and the C-reactive protein level was 0.6 mg per liter. Screening tests for antibodies to hepatitis B virus core antigen, hepatitis C virus, histone, and human immunodeficiency virus were negative, as were tests for Lyme disease antibodies, syphilis antibodies, rheumatoid factor, and varicella–zoster virus IgM; varicella–zoster virus IgG was detected. A test for SARS-CoV-2 RNA, an interferon- γ release assay for *Mycobacterium tuberculosis*, and serum and urine toxicology screens were also negative.

A lumbar puncture was performed. The cerebrospinal fluid (CSF) was clear, with an opening pressure of 14 cm of water. Gram's staining of the CSF showed rare mononuclear cells and no organisms. The CSF glucose level was 86 mg per deciliter (4.8 mmol per liter; reference range, 50 to 75 mg per deciliter [2.8 to 4.2 mmol per liter]), and the CSF protein level was 30 mg per deciliter (reference range, 5 to 55). There were no nucleated cells present.

On the second day, testing for antibodies to bartonella was negative. The levels of angiotensin-converting enzyme, C3, and C4 were normal, as were the results of serum free light-chain testing, serum protein electrophoresis, and urinalysis. The IgG4 level was 137.3 mg per deciliter (reference range, 3.9 to 86.4); the levels of other IgG subclasses were normal. Testing for lupus anticoagulant, anticardiolipin antibodies, and β_2 -glycoprotein 1 antibodies was negative, as was CSF testing for cryptococcal antigen and varicella–zoster virus IgM.

Dr. Ford: CT of the chest, performed after the administration of intravenous contrast material, showed no lymphadenopathy; there was hepatic steatosis and a 2-mm right apical nodule. MRI of the head and whole spine, performed after the administration of intravenous contrast material, showed asymmetric enhancement surrounding the intraorbital and intracanalicular segments of

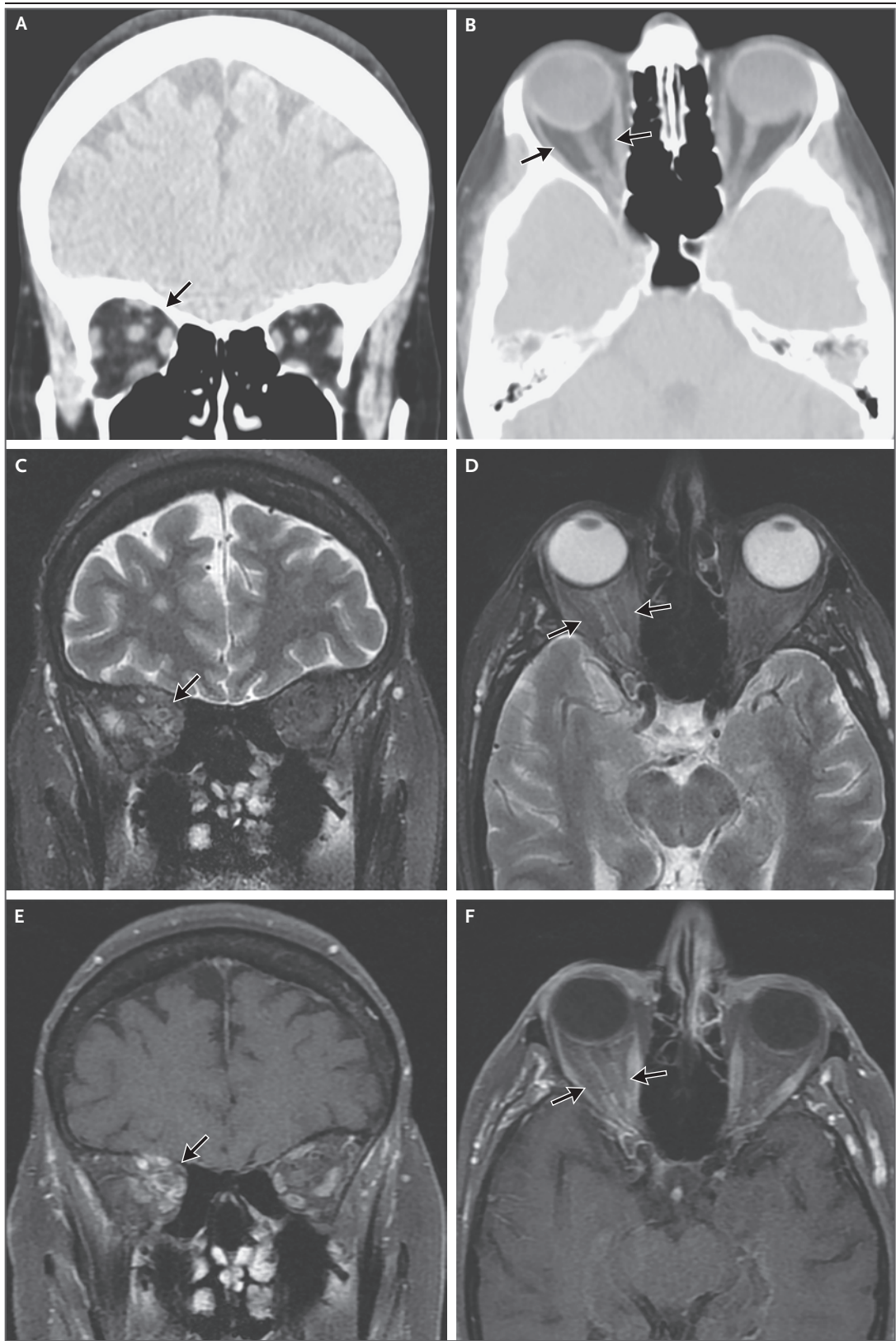


Figure 1 (facing page). Imaging Studies.

CT of the head was performed. Coronal and axial images (Panels A and B, respectively) show ill-defined stranding of the intraconal fat in the right orbit (arrows). MRI of the head, neck, and orbits was also performed. Coronal and axial T2-weighted, fat-suppressed images (Panels C and D, respectively) show high signal intensity in the right optic-nerve sheath with stranding of the intraconal fat (arrows). Coronal and axial T1-weighted, fat-suppressed images (Panels E and F, respectively), obtained after the administration of intravenous contrast material, show enhancement of the right optic-nerve sheath with ill-defined enhancement of the intraconal fat (arrows). The optic nerve itself is largely spared.

the right optic-nerve sheath and intraconal fat. No abnormal spinal cord enhancement was observed. Ultrasonography of the temporal arteries showed no halo sign.

Dr. Walton: A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Janeth M. Yin: This 67-year-old woman initially presented with a photosensitive rash that had developed while she was receiving hydrochlorothiazide; laboratory tests were positive for antinuclear antibodies (ANA), including antibodies against Sjögren's syndrome–related antigen A (SSA). Given that hydrochlorothiazide is known to be a common trigger of drug-induced subacute cutaneous lupus erythematosus,¹ one cannot help but consider the possibility that an autoimmune condition could have led to this patient's symptoms. However, a biopsy of the rash a few weeks after the initial presentation showed findings consistent with an allergic reaction that was presumed to be related to hydrochlorothiazide. Whether the rash was associated with the ocular symptoms was unclear at that time, but the results of the skin biopsy were not consistent with a dermatologic diagnosis of subacute cutaneous lupus erythematosus, autoimmune disease, or vasculitis.

The patient's course was complicated by acute sixth cranial nerve palsy that alternated from the left eye to the right eye. The initial left sixth cranial nerve palsy was self-limited and was presumed to be a microvascular event. Eight weeks later, a fundusoscopic examination revealed contralateral findings indicative of acute right sixth cranial nerve palsy with optic-nerve compromise. Results of repeat tests for inflammatory markers

(the erythrocyte sedimentation rate and C-reactive protein level) and basic laboratory tests were normal, and testing for Lyme disease was negative. MRI and magnetic resonance angiography of the head, neck, and orbits revealed a grossly spared optic nerve with evidence of edema involving the perineural and intraconal fat of the right eye.

When the anatomical trajectory of the sixth cranial nerve was tracked to localize the site of injury,² there was no evidence of hemorrhagic or ischemic microvascular disease at the brain stem in this patient, who had risk factors for small-vessel disease. She did not have meningeal enhancement at the subarachnoid portion of the nerve or CSF pleocytosis, which would be suggestive of inflammatory or infectious meningitis. Serologic and CSF testing for infectious diseases was also negative. There was no history of trauma or evidence of elevated intracranial pressure, and magnetic resonance angiography did not reveal any vascular anomalies or thrombosis at the cavernous sinus portion of the nerve trajectory. Edema of the optic perineural fat was seen at the point where the optic nerve traverses the optic canal into the orbit. Intraconal fat edema extended to the adjacent orbital fissure, resulting in injury of the traversing sixth cranial nerve and, consequently, right oculomotor dysfunction. Orbital apex syndrome, also called Jacod's syndrome, is characterized by vision impairment from optic neuropathy and ophthalmoplegia due to the involvement of the motor nerves in the anatomical region of the orbital apex, as was seen in this case.³

OPTIC PERINEURITIS

Optic perineuritis is characterized by inflammation restricted to the nerve sheath, whereas inflammation of the optic-nerve axons occurs with optic neuritis. Primary optic perineuritis is typically unilateral and idiopathic in nature. Bilateral idiopathic optic perineuritis is rare, and bilateral findings should prompt a search for an underlying infection such as syphilis, tuberculosis, herpes simplex virus infection, or herpes zoster; a systemic autoimmune disease such as antineutrophil cytoplasmic antibody (ANCA)–associated vasculitis, giant-cell arteritis, myelin oligodendrocyte glycoprotein (MOG) antibody disease, sarcoidosis, Graves' disease, Behçet's disease, or IgG4-related disease; or a malignant condition such as leukemia or primary or metastatic cancer.⁴

What caused the optic perineuritis in this patient? The distinction between primary optic perineuritis and optic neuritis with perineuritis is relevant because the cause, manifestations, treatment, and prognosis differ between the two conditions. Autoimmune demyelinating disorders that can cause optic neuritis include multiple sclerosis, neuromyelitis optica spectrum disorder (NMOSD), and MOG antibody disease. These diseases are unlikely to result in inflammation of the optic-nerve sheath without the presence of prominent parenchymal nerve enhancement. Swelling of the orbital soft tissue and oculomotor dysfunction are also atypical of these demyelinating disorders.⁵

Although Graves' disease, systemic lupus erythematosus (SLE), and MOG antibody disease can be associated with both optic perineuritis and optic neuritis, these autoimmune diseases are more frequently associated with optic perineuritis. This patient had autoimmune findings that included a remote history of Graves' disease with orbitopathy, a family history of vitiligo, and positive ANA and anti-SSA antibody tests.

Reactivation of thyroid eye disease, which can occur spontaneously or in association with SARS-CoV-2 vaccination, should be considered.⁶ The timing of her vaccinations against SARS-CoV-2 was relatively distant, and the extraocular muscle enlargement associated with thyroid eye disease was not present in this patient. Testing for common infections that can lead to perineuritis was unremarkable in this patient, and she had no major risk factors for infection. Optic perineuritis resulting from malignant infiltration, leptomeningeal disease, or a paraneoplastic syndrome would be a diagnostic alternative for which an invasive tissue biopsy could be considered for confirmation after a negative workup for autoimmune and autoinflammatory conditions.

SLE-ASSOCIATED OPTIC PERINEURITIS

SLE-associated optic perineuritis should be considered. This patient had a positive ANA test. However, other classic features of SLE were absent, and the results of other autoimmune-specific tests (e.g., tests for anti-double-stranded DNA, anti-Smith, and antiphospholipid antibodies) that would allow for the classification of SLE according to established criteria were negative in this patient.⁷

NEUROLOGIC SJÖGREN'S SYNDROME

A positive test for anti-SSA antibodies can be associated with Sjögren's syndrome, SLE, subacute cutaneous lupus erythematosus, or neonatal lupus. In addition, an increased frequency in the occurrence of a positive test for anti-aquaporin-4 (AQP4) antibodies has been described in patients with NMOSD.⁸ Sjögren's syndrome associated with optic perineuritis is rare, but cranial neuropathies are not an uncommon manifestation of neurologic Sjögren's syndrome. Although this patient had no history of sicca complex or extraglandular manifestations, which are findings typical of Sjögren's syndrome, central nervous system involvement may precede the diagnosis of primary Sjögren's syndrome.⁹ Longitudinal monitoring for Sjögren's syndrome should be performed, along with testing for anti-AQP4 and anti-MOG antibodies to rule out NMOSD and MOG antibody disease, respectively.

NEUROSARCOIDOSIS AND IGG4-RELATED DISEASE

Cranial neuropathy resulting from granulomatous invasion of one or more cranial nerves occurs in approximately 60% of patients with neurosarcoidosis. However, lung involvement is absent in only approximately 10% of patients with sarcoidosis.¹⁰ Optic neuropathy is seen in 5 to 9% of patients with IgG4-related disease with ophthalmic manifestations, and this diagnosis should be considered in this patient, given the mild elevation of the serum IgG4 level. Nonetheless, serum testing for IgG4 has a low sensitivity and specificity for the diagnosis of IgG4-related disease; the level is normal in 30 to 40% of patients.¹¹ Further evaluation for extrapulmonary sarcoidosis and for IgG4-related disease with other organ involvement, performed with the use of whole-body positron-emission tomography, can be considered to identify abnormalities for which a diagnostic biopsy may be indicated.

SMALL-VESSEL AND MEDIUM-VESSEL VASCULITIS

Optic perineuritis can result from vasculitis involving small and medium vessels that supply the orbital segment of the optic nerve, which can occur in the context of giant-cell arteritis or ANCA-associated vasculitis. These conditions are occasionally associated with optic perineuritis.^{4,5,12} The normal levels of acute-phase reactants and the absence of systemic inflammation would be unusual findings for giant-cell arteritis or

ANCA-associated vasculitis. A negative ultrasound of the temporal arteries makes the diagnosis of giant-cell arteritis less likely, and a normal erythrocyte sedimentation rate and C-reactive protein level would occur in only 0.8% of confirmed cases of temporal arteritis.¹³

Optic perineuritis has been increasingly recognized as a feature of granulomatosis with polyangiitis.^{5,14} Limited forms of granulomatosis with polyangiitis can manifest with neuro-ophthalmic findings and with minimal or no systemic findings, lacking components of the full triad.¹⁵ If other causes of optic perineuritis have been ruled out in the patient, and there is no low-risk site for a diagnostic biopsy, ANCA testing with reflex testing for antibodies to myeloperoxidase (MPO) and proteinase 3 (PR3) is a highly specific and accessible testing procedure with prompt results that could help to confirm a diagnosis of granulomatosis with polyangiitis in the presence of relevant clinical features.

DR. JANETH M. YINH'S DIAGNOSIS

Orbital apex syndrome due to antineutrophil cytoplasmic antibody-associated vasculitis with optic perineuritis.

DIAGNOSTIC TESTING

Dr. Reza Zonozi: The diagnostic test performed in this case was serum ANCA testing. ANCAs are autoantibodies directed against antigens in the cytoplasmic granules of neutrophils and monocytes. There are many target antigens, including bactericidal-permeability-increasing protein, lactoferrin, cathepsin G, elastase, MPO, and PR3.¹⁶ However, only ANCAs directed against MPO or PR3 are associated with systemic vasculitis.

ANCA testing includes two distinct tests: a microscopic examination with indirect immunofluorescence staining and an enzyme-linked immunosorbent assay (ELISA). Indirect immunofluorescence staining can detect ANCAs, but the findings are nonspecific. Positive indirect immunofluorescence staining can reveal ANCA with a cytoplasmic pattern (c-ANCA) or with a perinuclear pattern (p-ANCA). However, a positive test does not specify which autoantigen is being targeted. An antigen-specific ELISA for MPO and PR3 is essential to confirm the diagnosis.¹⁷ Moreover, the presence of ANAs can lead to an erroneous

interpretation of indirect immunofluorescence staining as positive for p-ANCA.¹⁸ This point further underscores the importance of the ELISA results. After this patient's serum was incubated with ethanol-fixed human neutrophils, indirect immunofluorescence staining was positive for p-ANCA. The ELISA was positive for antibodies to MPO (23 U; reference value, <2.8) and was negative for antibodies to PR3.

ANCA-associated vasculitis refers to systemic necrotizing vasculitis that predominantly affects small vessels and includes a spectrum of clinical phenotypes — namely, granulomatosis with polyangiitis, microscopic polyangiitis, eosinophilic granulomatosis with polyangiitis, and isolated pauci-immune necrotizing and crescentic glomerulonephritis (renal-limited vasculitis).¹⁹ The phenotypes can differ with respect to the predominant clinical features, the detection of granulomatous lesions on histologic examination, the ANCA serotype (in which PR3 antibodies are typically detected with granulomatosis with polyangiitis, and MPO antibodies are typically detected with microscopic polyangiitis), and the presence of asthma and eosinophilia; however, the phenotypes can have overlapping features.

A tissue biopsy is the preferred method for the diagnosis of vasculitis; however, an acceptable alternative is a positive test for ANCA to MPO or PR3 in patients who present with clinical features of vasculitis.²⁰ In a meta-analysis of the diagnostic value of ANCA testing, a combined ANCA testing system incorporating indirect immunofluorescence staining and the ELISA specific for MPO and PR3 had 99% specificity and 85% sensitivity.¹⁷ The usefulness of these findings depends on the pretest probability of vasculitis for any given clinical scenario. For example, in a patient with rapidly progressive glomerulonephritis, the probability of vasculitis before ANCA testing is approximately 50%.²¹ A positive test for ANCA to MPO or PR3 increases the probability of vasculitis to 99%.¹⁷

In this patient, the salient clinical features are bilateral, asynchronous sixth cranial nerve palsies and optic perineuritis. With some extrapolation from the literature, we can conservatively estimate a pretest probability of vasculitis of approximately 10%, with a possible increase to approximately 25% after other causes have been ruled out.^{5,22} After we factor in the positive MPO

ANCA test, the post-test probability of vasculitis is greater than 90%.¹⁷ This probability is further strengthened because the MPO ANCA level is unequivocally positive, relative to the cutoff value. This finding supports a diagnosis of ANCA-associated vasculitis.

PATHOLOGICAL DIAGNOSIS

Myeloperoxidase antineutrophil cytoplasmic antibody–associated vasculitis.

DISCUSSION OF MANAGEMENT

Dr. April M. Jorge: After the diagnosis of ANCA-associated vasculitis was established, we evaluated the patient to determine the extent of organ involvement and the severity of the disease. She had no evidence of glomerulonephritis, with bland urinalysis findings and normal renal function. The photosensitive rash on the forearms with the positive test for anti-SSA antibodies may have indicated drug-induced cutaneous lupus; the rash resolved with discontinuation of hydrochlorothiazide and was deemed to be unrelated to the ocular symptoms. The second rash was also suggestive of a drug reaction that was presumed to be related to hydrochlorothiazide and unrelated to the ocular symptoms. Overall, the disease was limited to right optic perineuritis and sixth cranial nerve palsy (as opposed to orbital inflammation as the cause of the diplopia), with no signs of other organ involvement. Therefore, treatment was initiated to target this limited but severe and potentially organ-threatening manifestation, with the goals of inducing remission and preventing organ damage while also minimizing adverse effects.

The patient had already received a dose of intravenous methylprednisolone. Treatment with high-dose oral prednisone was initiated, and within 3 days, the diplopia had resolved; the retro-orbital headache had already resolved before the initiation of this therapy. Treatment with the anti-CD20 monoclonal antibody rituximab was initiated for remission induction.^{23,24} This agent was chosen owing to the safety profile and evidence of noninferiority to cyclophosphamide and in accordance with the 2021 American College of Rheumatology–Vasculitis

Foundation guidelines for the management of ANCA-associated vasculitis.^{23,24} The patient adhered to a fixed-dose regimen of two intravenous infusions of rituximab given 2 weeks apart.

Additional treatment considerations included the glucocorticoid tapering regimen and the addition of the C5a receptor inhibitor avacopan. A tapering course of reduced-dose prednisone was prescribed; this decision was guided by the importance of minimizing the risk of glucocorticoid-related toxic effects²⁵ and was informed by the results of the PEXIVAS trial,²⁶ which showed a reduced-dose prednisone taper to be noninferior to the standard higher-dose prednisone taper in patients with ANCA-associated vasculitis.²³ Avacopan has been found to have glucocorticoid-sparing effects in the treatment of ANCA-associated vasculitis and was approved by the Food and Drug Administration in 2021 as an adjunctive treatment for this disease.^{27,28} The addition of avacopan to the treatment regimen was recommended for these reasons; however, this medication was not obtained, owing to the prohibitive cost. Therapeutic plasma exchange was not recommended, owing to a lack of evidence of benefit in patients with ANCA-associated vasculitis without renal involvement.^{26,29}

To reduce the risk of *Pneumocystis jirovecii* pneumonia due to intensive immunosuppression, the patient received trimethoprim–sulfamethoxazole prophylaxis during treatment with moderate-to-high doses of prednisone. In addition, she received calcium and vitamin D supplementation and bisphosphonate alendronate to reduce the risk of glucocorticoid-induced osteoporosis. Several adverse effects related to glucocorticoid therapy occurred in this patient, including insomnia, trouble concentrating, hyperphagia, facial swelling, tremor, and new atrial fibrillation with rapid ventricular response.

Four months after the patient started treatment for ANCA-associated vasculitis, results of MRI of the orbits and an ophthalmologic examination were normal. The glucocorticoid taper was nearly complete, her symptoms had remained in remission, and ELISA testing showed MPO ANCA at a level of 2.3 U. A plan was made for the patient to receive maintenance therapy with rituximab starting at the 6-month follow-up visit.

FINAL DIAGNOSIS

Myeloperoxidase antineutrophil cytoplasmic antibody-associated vasculitis.

This case was presented at Rheumatology Grand Rounds. Disclosure forms provided by the authors are available with the full text of this article at NEJM.org. We thank Dr. Marcy Bolster for assistance with preparation of the case presentation.

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